

WET SCRUBBER LNS-SC-WET

Industrial Gas Cleaning &
Odour Control System



CLEANER
ENVIRONMENT



RELIABLE
OPERATION



PROVEN
TECHNOLOGY

Vapour Inlet
(Contaminated Air)

Clean Air Outlet
to Atmosphere

Circulation Line
to Internal Spray
System

CLEAN
PROCESS
CLEANER
AIR
BRIGHTER
TOMORROW

Electrical
Control Panel

Level Indicator
with Controller

Fresh Water Feed
(with Valve)

Circulation Pump

Drain Connection
(with Valve)

*SOLUTIONS FOR A
CLEANER TOMORROW*

INDUSTRIAL WET SCRUBBER SYSTEM

1. PROCESS OVERVIEW

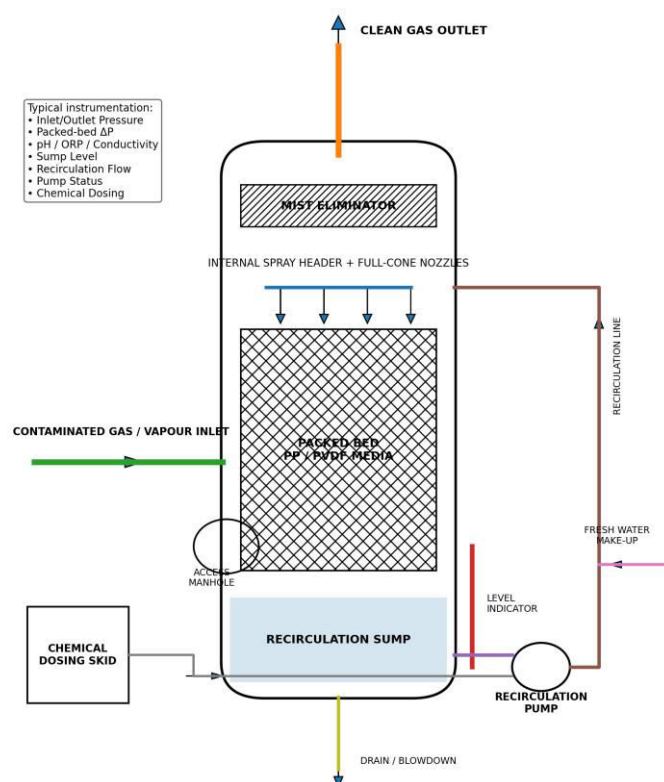
The **LINESIG LNS-SC-WET** Industrial Wet Scrubber is a high-efficiency vertical packed-bed gas-cleaning system designed for continuous removal of water-soluble gases, acidic or alkaline vapours, odorous compounds, and selected fine aerosols from industrial exhaust streams. Contaminated gas enters the lower section of the scrubber and flows upward counter-current to a continuously recirculated scrubbing liquid. Target contaminants are transferred from the gas phase into the liquid film on the high-surface-area packing and are neutralized, oxidized, absorbed, or otherwise conditioned by the selected reagent chemistry.

The treated gas passes through an integral mist eliminator to minimize liquid carryover before discharge to the stack. The recirculation sump, pump system, fresh-water make-up, blowdown arrangement, chemical dosing, and automated pH/ORP/conductivity control are integrated into one engineered package. Unlike a biological scrubber, the LNS-SC-WET relies primarily on physical absorption and controlled chemical reaction rather than microbial degradation.

2. INDUSTRIAL APPLICATIONS

- Chemical & Petrochemical Plants:**
HCl, SO₂, HF, acid mist, caustic fumes, solvent-water soluble vapours, reactor and storage-tank vent treatment.
- Fertilizer & Ammonia Facilities:**
NH₃, alkaline vapours, ammonium compounds, acid neutralization duties and process-vent polishing.
- Metal Processing / Pickling / Galvanizing:**
Hydrochloric, sulfuric and nitric acid fumes, metal-treatment vapours and rinse-line exhaust.
- Wastewater / Sewage Treatment:**
H₂S, NH₃, mercaptans and nuisance odours from headworks, tanks, sludge handling and enclosed process areas.
- Food, Rendering & Fermentation:**
Organic odours, ammonia, amines, soluble VOCs and process exhaust from cookers, dryers and fermentation systems.
- Pulp & Paper / Textile / Dyeing:**
Reduced sulfur compounds, chlorine-bearing vapours, acid/alkaline process fumes and odour control.
- Battery / Chemical Storage / Recycling:**
Acid mist and corrosive vapours from charging, transfer, treatment and recovery processes.

LNS-SC-WET — VERTICAL PACKED-BED WET SCRUBBER PROCESS SCHEMATIC



PROCESS SCHEMATIC — LNS-SC-WET

Process Sequence:

contaminated gas → inlet connection / optional damper → lower gas-distribution zone → packed absorption section → internal liquid distributor and spray system → mist eliminator → clean-gas outlet. Recirculating liquid is drawn from the sump, pumped to the internal spray header, chemically conditioned as required, and partially blown down to control dissolved solids.



3. TECHNICAL & PERFORMANCE SPECIFICATIONS:

Parameter	Nominal Value / Design Criterion	Engineering Comments
Nominal Volumetric Gas Flow Rate	10,000 m ³ /h (reference model basis)	System is scalable; final diameter and blower duty depend on gas loading and pressure drop.
Gas Velocity Through Packing	Typically 1.5 – 2.5 m/s	Selected to balance mass transfer, pressure drop and entrainment control.
Liquid-to-Gas Ratio	Typically 3 – 8 L/m ³	Adjusted for pollutant solubility, stoichiometry and wetting requirement.
Packed-Bed Pressure Drop	Typically < 300 Pa/m clean bed	Final value depends on packing type, gas velocity and liquid rate.
Recirculation Liquid pH	Application dependent	Commonly controlled by acid/alkali dosing for target chemistry.
Target Removal Efficiency	Typically 90 – 99%+ for suitable soluble/reactive contaminants	Actual performance depends on inlet load, reagent chemistry, contact time and equilibrium.
Mist Carryover	Designed for high-efficiency droplet removal	Demister selected for final gas velocity and droplet loading.

4. ABSORBER COLUMN MECHANICS

Shell Construction:

FRP/SS/CS with Vinyl Ester resin liner, PP, PVC, PVDF, stainless steel, or coated carbon steel may be selected according to corrosivity, temperature, pressure and site conditions.

Packing Elements:

High-efficiency random or structured PP/PVDF/SS packing selected to provide high specific surface area, low pressure drops and uniform liquid distribution.

Liquid Distribution Network:

Removable corrosion-resistant internal headers fitted with full-cone or wide-angle spray nozzles to achieve complete cross-sectional wetting.

Mist Eliminator:

High-efficiency PP/PVDF chevron or mesh-pad demister installed above the spray section to control entrained droplets before discharge.

Component	Mechanical Specification & Design Features
Shell Material (MOC)	FRP/Vinyl Ester, PP, PVC, PVDF, SS316L or coated CS — project specific
Packing Medium	PP/PVDF random or structured packing; type and surface area selected by duty
Liquid Distributor	Removable corrosion-resistant header with full-cone spray nozzles
Mist Eliminator	PP/PVDF chevron or mesh-pad type; selected for gas velocity and droplet loading
Access	Manhole / packing fill and removal access / inspection ports as required



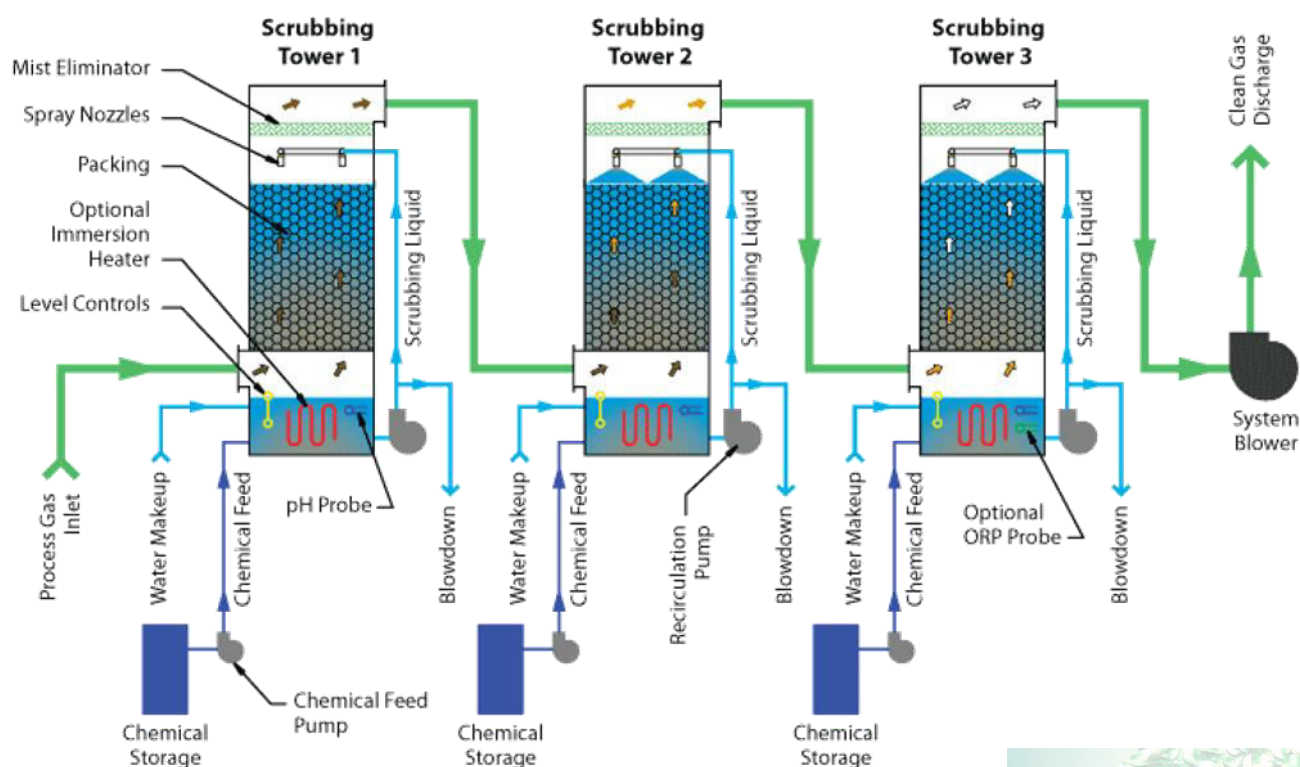
5. CHEMICAL ABSORPTION & REACTION PRINCIPLES

Wet scrubbing performance is obtained by combining gas-to-liquid mass transfer with the chemistry most suitable for the pollutant. The recirculation liquor is continuously monitored and conditioned so that the driving force for absorption remains high.

Pollutant / Service	Typical Scrubbing Reagent	Primary Mechanism
HCl / HF / SO ₂ / acidic gases	NaOH, Na ₂ CO ₃ or alkaline liquor	Neutralization and absorption
NH ₃ / alkaline vapours	Dilute H ₂ SO ₄ or other approved acid	Acid absorption / salt formation
H ₂ S / mercaptans	NaOH + NaOCl or other oxidizing chemistry	Alkaline absorption + oxidation
Chlorine / Cl ₂	NaOH / reducing chemistry as process requires	Reactive absorption
Odorous organics / soluble VOCs	Water or chemically conditioned water	Physical absorption / reaction
Acid mist / soluble aerosols	Water / neutralizing liquor	Inertial capture + dissolution

6. RECIRCULATION LIQUOR MANAGEMENT

- Automatic fresh-water make-up maintains sump operating level.
- Continuous or batch chemical dosing maintains the required pH/ORP window.
- Conductivity or total dissolved solids may be used to control blowdown.
- Low recirculation flow shall interlock chemical dosing to prevent concentrated chemical accumulation.
- Sump agitation or recirculation velocity shall prevent solids settlement where salts or precipitates may form.
- Blowdown shall be routed to the client's approved effluent-neutralization or wastewater-treatment system.



7. PROCESS GAS BLOWER SUBSYSTEM

Configuration & Material: Heavy-duty centrifugal process blower, direct or belt driven, selected for the complete system static pressure including ductwork, packed-bed resistance, mist eliminator, dampers and stack losses. Wetted/corrosive gas components may use FRP, PP, coated steel or other suitable construction.

Operational Basis: Reference package basis is 10,000 m³/h. Final blower static pressure is determined from project duct routing and scrubber pressure-drop calculation rather than assumed as a fixed value.

Equipment Subsystem	Detailed Engineering Parameters
Process Gas Blower Type	Industrial centrifugal blower, direct or belt driven
Reference Gas Flow	10,000 m ³ /h continuous — scalable
Static Pressure	Project calculated; typically sized with operating/fouling margin
Construction	FRP / PP / coated steel / SS according to gas chemistry
Motor / Control	TEFC motor; VFD recommended for flow or header-pressure control
Scrubbing Water Pump	Horizontal end-suction centrifugal or vertical sump pump
Reference Recirculation Flow	Application dependent; preliminary 35 – 45 m ³ /h for 10,000 m ³ /h reference duty
Reference TDH	Typically 2.5 – 3.5 bar, confirmed by hydraulic calculation
Redundancy	1 Duty + 1 Standby recommended for critical 24/7 service

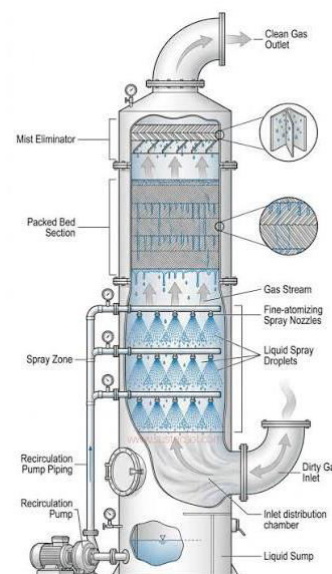
8. SCRUBBING WATER RECIRCULATION PUMP SUBSYSTEM

Design Architecture:

Pump(s) draw recirculation liquor from the scrubber sump and deliver it through the external riser to the internal spray header. For continuous critical services, a 1 Duty + 1 Standby arrangement with automatic changeover is recommended.

Materials & Seals:

PP, PVDF, FRP-lined or corrosion-resistant metallic construction may be selected. Mechanical seals shall be chosen for reagent chemistry and suspended-solids content; sealless magnetic-drive options may be considered for highly corrosive duties.



Parameter	Reference / Design Basis
Pump Type	Horizontal end-suction centrifugal or vertical sump pump
Flow	Project-specific; preliminary 35 – 45 m ³ /h at 10,000 m ³ /h gas basis
TDH	Project-specific; typically, 25 – 35 m water column
MOC	PP / PVDF / FRP / SS316L or approved equivalent
Control	DOL or VFD; flow switch/transmitter used for permissive
Redundancy	1D + 1S for critical service

9. CHEMICAL INJECTION SYSTEM & REAGENT MATRIX (OPTIONAL)

High-precision electromagnetic or motor-driven diaphragm dosing pumps shall inject reagent into the recirculation circuit or sump. Dosing rate is automatically modulated from pH, ORP, conductivity, or process-specific analyzer feedback.



Reagent / Function	Typical Use
Sodium Hydroxide (NaOH)	Neutralization of acidic gases such as HCl and SO ₂
Sulfuric Acid (H ₂ SO ₄)	Absorption/neutralization of ammonia or alkaline vapours
Sodium Hypochlorite (NaOCl)	Oxidation of H ₂ S and selected reduced sulfur compounds
Hydrogen Peroxide (H ₂ O ₂)	Alternative oxidation chemistry where appropriate
Antiscalant / Dispersant	Optional, where salt precipitation or scaling is expected

10. INTEGRATED PROCESS CONTROLS & INSTRUMENTATION

The system may be supplied with a centralized PLC-based control panel in an IP65/NEMA 4X enclosure. The exact platform is project specific; Siemens S7-1200/1500 or equivalent industrial PLC architecture may be used.

Control Loop	Automation & Instrument Strategy
Sump Level Control	Level transmitter / switches command fresh-water make-up valve and protect recirculation pump.
pH Control Loop	Online pH analyzer modulates acid or alkali dosing to maintain selected operating band.
ORP Control Loop	Optional for oxidizing duties such as H ₂ S / mercaptan treatment.
Conductivity / Blowdown	Conductivity or TDS signal initiates blowdown to control dissolved solids accumulation.
Recirculation Flow	Flow transmitter / low-flow switch proves spray circulation and interlocks chemical dosing.
Packed-Bed Differential Pressure	DP transmitter monitors fouling, scaling or media blockage.
Gas Blower Control	VFD control by flow, duct pressure or process demand.
Alarm / Trip	Low level, low flow, high DP, pump trip, chemical tank low level and analyzer fault.

11. TYPICAL INSTRUMENTATION

- PT / PI on inlet and outlet gas headers as required.
- PDT across packed bed and mist eliminator.
- LT with LSHH / LSL on scrubber sump.
- FT / FSL on recirculation discharge.
- pH analyzer as standard for neutralization duties.
- ORP analyzer for oxidation chemistry where required.
- Conductivity transmitter for blowdown control.
- Pressure gauge on pump discharge.
- Chemical tank level switches and dosing pump run/fault feedback.
- Local E-Stop, HOA/Auto selector and PLC/HMI diagnostics.

12. SYSTEM CONFIGURATION

The vertical packed-bed wet scrubber provides a compact footprint while allowing long gas–liquid contact length. Counter-current flow maximizes mass-transfer driving force and enables high removal efficiency for soluble and chemically reactive contaminants. The system can be supplied as a single-stage neutralization scrubber or as a multi-stage package where incompatible chemistries must be separated.

Configuration	Recommended Application
Single-Stage Water Scrubber	Highly water-soluble, low-reactivity vapours; polishing duties
Single-Stage Alkaline Scrubber	HCl, SO ₂ , HF and other acidic gases
Single-Stage Acid Scrubber	NH ₃ and alkaline vapours
Alkaline + Oxidation Scrubber	H ₂ S / mercaptans / selected odours
Two-Stage Scrubber	Mixed acidic/alkaline contaminants or high inlet load requiring sequential chemistry
Venturi + Packed Bed	Particulate/aerosol pre-capture followed by soluble-gas absorption

13. DESIGN SELECTION INPUTS

- Gas flow: minimum / normal / maximum.
- Gas temperature, pressure, humidity and dust/aerosol loading.
- Contaminant composition and concentration at normal and maximum conditions.
- Required outlet limit / removal efficiency.
- Available water quality and wastewater-disposal constraints.
- Selected reagent availability, concentration and storage philosophy.
- Corrosion requirements, hazardous-area classification and site ambient conditions.
- Duct routing / stack height / allowable system pressure drop.

14. STANDARD PACKAGE COMPONENTS

Item	Description
A	Scrubber body / packed absorption column
B	Clean-gas outlet / stack connection
C	Mist eliminator
D	Mist-eliminator access / inspection opening
E	Internal recirculation liquid distribution nozzle system
F	External recirculation piping / riser
G	Recirculation pump(s)
H	Optional basket strainer / duplex strainer
I	Liquid flow meter / low-flow protection
J	Packed mass-transfer media
K	Packing support grid
L	Contaminated gas inlet
M	Recirculation sump
N	Sump level-control system
O	Packing removal / access manhole
P	Packing fill / inspection door
Q	Fresh-water make-up valve
R	Chemical dosing skid and reagent tank(s)
S	Blowdown / drain connection
T	Electrical / PLC control panel



CLEANER INDUSTRY. HEALTHIER AIR. GREENER TOMORROW.

Advanced wet-scrubber solutions for responsible industrial emission and waste-air treatment.